NAFLAN MOHAMED

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PROFESSIONAL SUMMARY

AI Engineer focused on LLM applications, Retrieval-Augmented Generation (RAG), and agentic AI systems. Hands-on experience building document retrieval pipelines, embedding and vector search systems, multi-step LangGraph agents with tool calling and MCP, and end-to-end AI backends using Python, FastAPI, LangChain, LangGraph, and PostgreSQL/pgvector. Comfortable owning an AI feature from data ingestion through retrieval, generation, and deployment, with full-stack ability to ship the surrounding product experience.

TECHNICAL SKILLS

- **AI / LLM Engineering** — LLM Applications, Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, LangGraph, Prompt Engineering, Embeddings, Vector Search, Semantic Search, Tool Calling, MCP (Model Context Protocol), Gemini API, Hugging Face
- **Programming** — Python, JavaScript
- **Backend** — FastAPI, Django, REST APIs, Asynchronous Processing, PostgreSQL, pgvector, Pinecone, ChromaDB, Supabase
- **Supporting / Full-Stack** — React.js, Node.js, Docker, Postman, Git, GitHub, Vercel

WORK EXPERIENCE

NETSOL Technologies | *AI/ML Intern* — July 2026 - Present

- Build LLM application features end to end, including document chunking, embedding generation, vector retrieval, and context-aware generation for internal RAG workflows.
- Work daily with Python, FastAPI, LangChain, LangGraph Gemini API, REST APIs, and vector databases to develop and debug production LLM pipelines.
- Apply prompt engineering, retrieval optimization, asynchronous processing, and structured error handling to improve reliability of AI workflows.
- Extend these skills in a personal agentic AI project using LangGraph, RAG, tool calling, MCP, document retrieval, web search, and calendar/email integrations.

Fluxxion | *MERN Stack Developer Intern* — Jul 2025 - Aug 2025

- Refactored React component architecture using memoization and lazy-loading patterns, achieving a 40% improvement in Lighthouse performance scores and reducing reported UI defects by 30% within the first month.
- Designed and shipped a modular Daily Goals Tracker feature from scratch, integrated into the core product with full test coverage; adopted by the product team as a reusable pattern for future modules.
- Restructured MongoDB aggregation pipelines and added field-level indexing on high-traffic endpoints, reducing average API response latency by an estimated 35% under load.

AICP | *Web Developer Intern* — Mar 2024 - Jun 2024

- Developed and maintained 5+ production web applications in HTML/CSS/JavaScript; implemented client-side validation that reduced form submission errors by 60%.
- Refactored frontend codebases across 3 projects for full mobile responsiveness, eliminating all major layout breakages on tested devices and screen sizes.
- Integrated third-party APIs and dynamic content feeds into live applications, extending feature sets without increasing backend complexity or server load.

PROJECTS

AI ChatAgent Workspace

Python | LangGraph | Gemini | RAG | MCP | FastAPI | React | PostgreSQL | Supabase | Redis [GitHub Repo]

- Built a full-stack agentic AI workspace supporting 5+ tool categories, including web search, email, calendar actions, document retrieval, and URL analysis.
- Designed a LangGraph agent workflow to coordinate LLM reasoning, tool calling, RAG retrieval, and multi-step task execution.
- Implemented an end-to-end RAG pipeline covering document upload, parsing, chunking, embeddings, vector retrieval, and context-aware response generation.
- Integrated authentication, PostgreSQL/Supabase persistence, and Redis-based infrastructure to support the application.

Yaloo - AI-Powered Cultural Tourism Platform (FYP)

Flutter | Django | Supabase/PostgreSQL | RAG | Embedding | pgvector | SentenceTransformers | Gemini | Redis

- Architected and developed a cross-platform tourism platform connecting tourists with verified local guides and homestay hosts, tested 150+ active users across tourist, guide/host, and admin roles through a single Flutter codebase.
- Engineered an AI recommendation system using Sentence Transformers embeddings, pgvector, HNSW indexing, and hybrid scoring across guides, stays, and activities, combining semantic similarity with budget and activity preferences to deliver personalized recommendations.
- Built an intent-routed RAG chatbot powered by Gemini, implementing history-aware context, query validation, and multi-tier fallback handling; validated the system against 50+ test queries to improve retrieval reliability and reduce unnecessary API usage.
- Designed the backend and data architecture with Django REST Framework and Supabase/PostgreSQL, including booking, availability, reviews, chat, authentication, and 17 database-driven webhook workflows that automatically synchronize entity embeddings when platform data changes.

DocChat - AI Document Question-Answering Platform

Next.js | FastAPI | Python | RAG | LangChain | Gemini API | ChromaDB Link Demo

- Engineered and deployed an end-to-end RAG application that enables users to upload PDF, DOCX, and TXT documents and interact with them through context-aware AI conversations with source citations.
- Built the complete document-to-answer pipeline, including document parsing, recursive text chunking, embedding generation, vector storage, semantic retrieval, context construction, and Gemini-powered response generation using LangChain and ChromaDB.
- Developed a production-style FastAPI backend with asynchronous processing and streaming responses, integrating document ingestion, retrieval, chat sessions, and API services to deliver responsive document-based conversations.
- Implemented retrieval-grounded responses that constrain answers to relevant document context, improving answer reliability while reducing irrelevant LLM responses and unnecessary context passed to the model.

EDUCATION

BS (Hons) in Computer Science

Dec 2022 - June 2026

Government College University, Lahore, Pakistan

CERTIFICATIONS

- **Meta Full Stack Developer: Front-End & Back-End from Scratch** - Meta — Dec 2025
- **MERN Stack Front to Back: Full Stack React, Redux & Node.js** - Packt — Dec 2025
- **MERN eCommerce from Scratch** - Packt — Jul 2025